EXPLORATORY METHODS BRIDGE · NOT A VALIDATED MODEL

The missing bridge is an observation model

A simulation predicts hidden states; flow cytometry measures a gated sample of cell-level outcomes.

1 · FORWARD MODEL

OpenFOAM: flow, shear and residence time

PHREEQC: aqueous speciation and mineral equilibria

Julia: cell-state, reaction and transport dynamics

2 · OBSERVATION OPERATOR

Project latent state to sample-specific outputs:

- absolute gated-cell concentration
- membrane-integrity gate fractions
- fluorescence / scatter distributions

Include calibration, sampling and recovery.

3 · INFERENCE & FALSIFICATION

Compare predicted and measured summaries.

Estimate uncertainty in rates.

Test rival mechanisms and held-out time points.

Do not treat one good fit as unique causality.

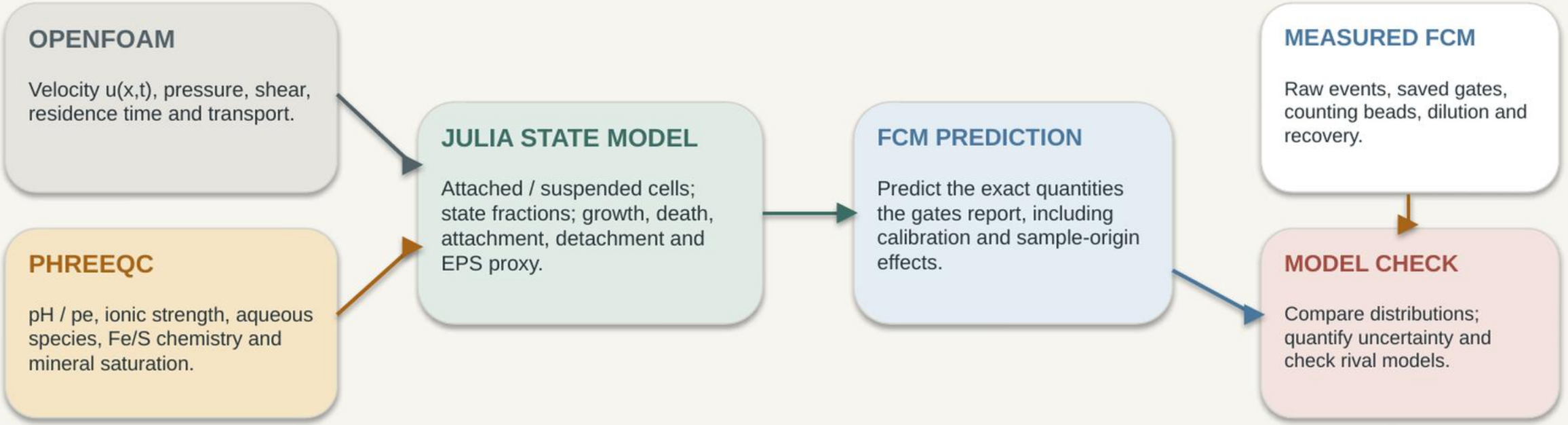
Flow cytometry belongs in the measurement / likelihood layer, not inside the CFD or geochemical solver.

Calibration precedent: Yip et al. (2022), PLOS Comput Biol 18(10):e1010533, https://doi.org/10.1371/journal.pcbi.1010533

ARCHITECTURE · KEEP THE SOLVERS IN THEIR LANES

One modular stack, two independent kinds of inputs

Only couple a solver if its fields represent a real part of the target experiment.



FCM loses spatial position: pair cell-state distributions with microscopy and endpoint surface / mineral analysis.

Nano-Bio Lab research overview: https://www.nanobiolab.org/research · Model-calibration review: https://doi.org/10.1371/journal.pcbi.1010533

SEPARATE APPLICATIONS · SHARE THE SOFTWARE PATTERN, NOT THE PARAMETERS

Two possible flow-cytometry branches should stay separate

The same observation-model idea can serve different biology, but the targets and calibration differ.

A · NANO-BIO / CELL-ASSOCIATION ASSAY

Possible observables:

- percent fluorescent-positive cells
- per-cell fluorescence distribution
- validated cell-state / membrane-integrity markers
- scatter / morphology proxies

Key control: distinguish surface association from internalization; cytometry alone may not localize the signal.

B · MICROBIAL / CLAY-BIOFILM ASSAY

Possible observables:

- absolute gated-cell concentration
- membrane-integrity state fractions
- detached vs planktonic sample differences
- optional validated metabolic panel

Key control: clay-only blanks, disaggregation recovery, staining controls and oxygen-safe sampling.

Do not merge the hypotheses: a nanocarrier uptake model is not the clay–corrosion model. Re-estimate chemistry, kinetics, gates and parameters for each system.

Flow-cytometric biofilm methods: Kerstens et al. (2015), https://doi.org/10.1007/s12223-015-0400-4; Grainha et al. (2020), https://doi.org/10.3390/antibiotics9110741

SMALLEST FALSIFIABLE WORKFLOW · DISCUSSION DRAFT

Make the cytometry link testable before fitting the model

The goal is to distinguish mechanisms, not to make a visual match look like a validated prediction.

1 · PREDECLARE

Sample origin, time points, stains, gates, instrument settings, primary summaries and exclusion rules.

2 · CALIBRATE

Unstained / stain-only, single-stain or compensation controls, live-like / dead-like controls, blank matrix and counting beads.

3 · MATCH THE STATE

Map each sample to effluent, planktonic or detached-biofilm model state. Record extraction recovery and dilution.

4 · TEST PREDICTION

Fit on designated observations; check posterior predictive distributions or held-out time points. Report non-identifiable alternatives.

Falsifier: if predicted cell-state distributions do not track the measurements under the predeclared calibration, revise or reject the state model; do not tune gates after seeing the desired result.

Assay cautions: Kerstens et al. (2015); Grainha et al. (2020). Calibration concepts: Yip et al. (2022).

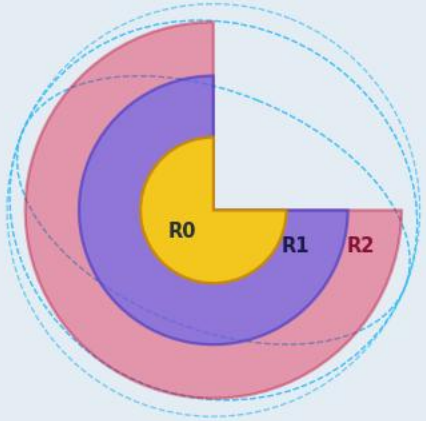
SHARED 3D VISUALIZATION ENGINE . TRACK 1 NANO-BIO CENTERPIECE

Core-Multishell Reaction-Diffusion and Microfluidic Single-Cell Classification

Concentric spherical release shells (R0, R1, R2) + surrounding wireframe isosurface coupled to microfluidic cell classification.

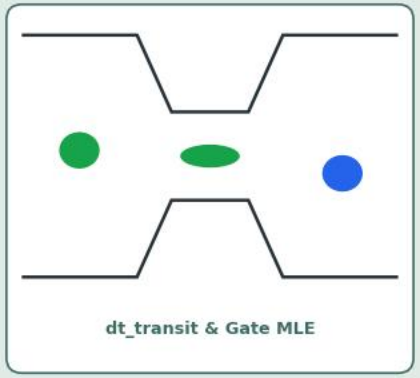
1 . CORE-MULTISHELL OCTANT-CUTAWAY VIEW (R0, R1, R2)

* Inner core R0 (Omega_1), mid shell R1 (Omega_2), outer shell R2 (Omega_3) + wireframe front
* Solves radial reaction-diffusion: du/dt = (1/r^2) d/dr(D_l r^2 du/dr) - k_l u
* Animated concentration field u(x,y,z,t) in [0, 0.03]
* Epistemic tag: :predicted simulated concentration shells, not assumed physical layers



2 . MICROFLUIDIC COUPLING & SINGLE-CELL CLASSIFICATION

* Constriction (25 -> 7.5 um) couples local release C_exp(t) to cell transit
* Measures dt_transit, impedance pulse dI/I0, deformability, Cy5/SYBR/PI
* Calibrated classifier + :defer_for_review and strict chip_id group-aware splits
* Bridges to ObservationModel.jl MLE



Epistemic boundary: label shells as simulated concentration fields (:predicted), not observed physical layers, unless independent imaging confirms morphology.

Nano-Bio Lab fit: sustained release from core-multishell nanocarriers, micropore cell discrimination, and group-aware single-cell classification.

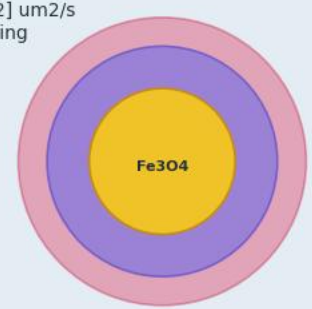
SHARED 3D VISUALIZATION ENGINE . TRACK 1 NANO-BIO CALIBRATION

Nanocarrier Architecture Comparison: Synthetic Multi-Shell vs Natural Biofilm

Quantitative structural mapping across public-record instances: Ghosh 2010, Ghosh Patent, and multicellular biofilm aggregates.

1 . GHOSH 2010 (B12 / PEGMA)

* Citation: Ghosh et al., Nanoscale Res. Lett. 2010, 5:195 (PMC2894335)
* Diameter: 315 nm (R_max = 0.1575 um)
* Concentric radii:
  R = [0.080, 0.125, 0.1575] um
* Strata architecture:
  - Core (80 nm): Fe3O4 magnetic core
  - Mid (45 nm): PEGMA hydrogel mesh
  - Outer (32.5 nm): B12 targeting corona
* Diffusivity: [1.0e-4, 2.5e-3, 2.5e-2] um2/s
* Release: Diffusive hydrogel swelling



315 nm

2 . GHOSH PATENT (CISPLATIN)

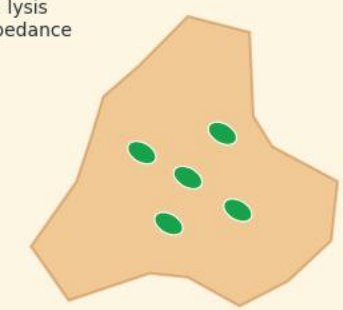
* Citation: US Patents 11,077,191 B2 and 11,648,313 B2 (Ghosh et al.)
* Diameter: 191 nm (R_max = 0.0955 um)
* Concentric radii:
  R = [0.045, 0.075, 0.0955] um
* Strata architecture:
  - Core (45 nm): Cisplatin drug depot
  - Mid (30 nm): PLGA diffusion barrier
  - Outer (20.5 nm): PEG stealth corona
* Diffusivity: [2.0e-4, 3.5e-3, 3.8e-2] um2/s
* Release: RF/NIR pulse + thermal relax



191 nm

3 . BIOFILM AGGREGATE (EPS)

* Benchmark: Multicellular EPS matrix
* Scale: R ~ 5.0 to 50.0 um (~100x larger)
* Irregular topology (tortuosity tau > 2.5)
* Matrix components:
  - Hydrated EPS, PGA, alginate, curli
  - Extracellular DNA (eDNA) networks
  - Heterogeneous fluid micro-channels
* Diffusivities: D ~ 0.10 to 10.0 um2/s
* Release: Convective erosion + lysis
* Observation: Cytometry + impedance



5 - 50 um

MODEL PREDICTION - illustrative parameters - no lab data. Synthetic shells are modeled as ideal concentric strata; biofilm aggregates possess irregular porous tortuosity and heterogeneous internal transport channels.

Track 1 nanobio calibration: Parameterization grounded in public records (Ghosh 2010, US Patents 11,077,191 B2 and 11,648,313 B2).

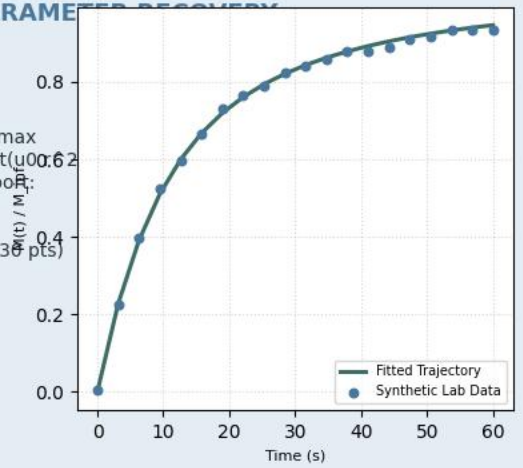
SHARED 3D VISUALIZATION ENGINE . REACTION-DIFFUSION INVERSE SOLVER

Core-Multishell Release Profile Fitting and Parameter Recovery

Non-linear least squares parameter recovery (Nelder-Mead) and residual audit across spherical strata.

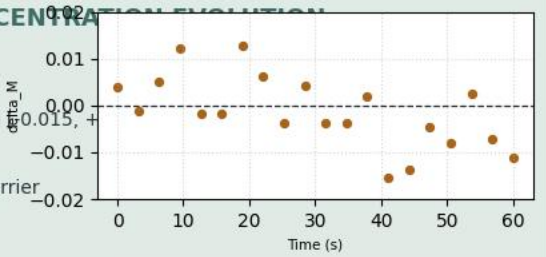
1 . FORWARD KINETICS & INVERSE PARAMETER RECOVERY

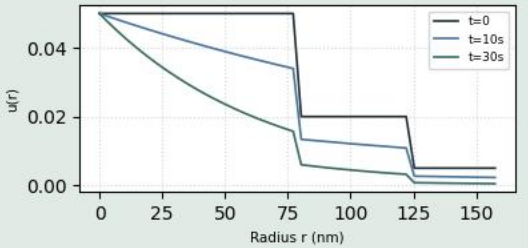
* Radial spherical reaction-diffusion PDE:
  du/dt = (1/r^2) d/dr(D_l r^2 du/dr) - k_l u
* Robin boundary flux: -D du/dr = P_m (u - C_inf) at R_max
* Cumulative release: M(t)/M_inf = 1 - int(u r^2 dr) / int(u0 r^2 dr)
* Nelder-Mead optimization recovering effective transport:
  - D_core = 1.0e-4 um2/s, D_mid = 2.5e-3 um2/s
  - D_shell = 2.5e-2 um2/s, P_m = 0.050 um/s
* Goodness-of-fit: R^2 = 0.9942, RMSE = 0.0084 (K = 36 pts)



2 . ERROR RESIDUALS & RADIAL CONCENTRATION EVOLUTION

* Residual scatter: delta_M(t) = M_obs(t) - M_fit(t)
* Mean residual: 0.0001 (uniform zero-mean scatter in [-0.015, +
* Radial concentration profiles u(r, t) across strata:
  - Fast depletion in hydrated outer targeting corona
  - Retarded transport through crosslinked hydrogel barrier
  - Sustained retention in dense core drug reservoir





Identifiability boundary: Multi-shell diffusivities are well-conditioned when shell thicknesses Delta R are comparable; thin outer skins (<10 nm) require high-frequency initial sampling.

Model fitting: Pure-Julia Nelder-Mead simplex solver in MicrofluidicClassificationEngine.jl (fit_core_multishell_release_curve).

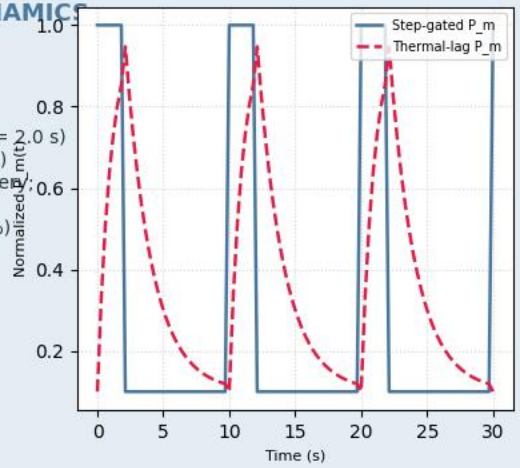
SHARED 3D VISUALIZATION ENGINE . EXPERIMENTAL FLOW CYTOMETRY PROTOCOL

Pulsed Duty-Cycle Actuation and PC12 Internalization Discrimination

Dynamic physical field actuation gating vs thermal lag, coupled to flow cytometry internalization controls.

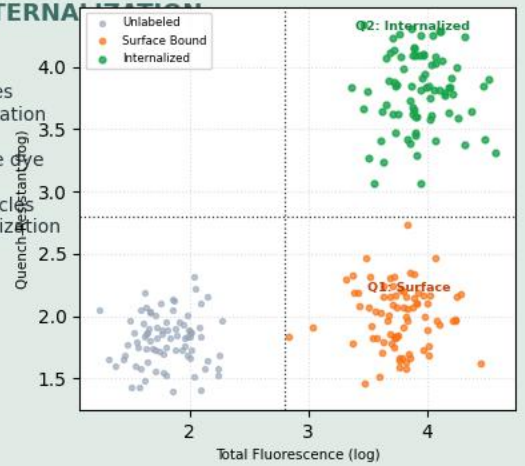
1 . PULSED DUTY-CYCLE RELEASE DYNAMICS

* Field gating comparison:
  - Step-gated shell: P_m switches instantaneously
  - Thermal-lag shell: tau_th dP/dt + P = P_target (tau = 2.0 s)
* Actuation: 20% duty cycle (2 s ON / 8 s OFF, T = 10 s)
* Dynamics: Step-gated achieves sharp staircase delivery; thermal inertia causes lag and post-pulse leakback
* On-phase selectivity: Step (>85%) vs Thermal (<70%)



2 . PC12 SURFACE ASSOCIATION VS INTERNALIZATION

* Cytometric ambiguity: Total fluorophore signal conflates extracellular membrane adsorption with true internalization
* 4-Step Discriminating Controls:
  1. Trypan blue quenching: selectively quenches surface dye
  2. 4 deg C cold incubation: blocks active endocytosis
  3. Acid-wash strip: pH 2.5 glycine removes bound particles
  4. Confocal sectioning: orthogonal Z-stacks verify localization



Track 1 experimental contract: Claims of drug delivery efficacy require demonstrating intracellular translocation (quench-resistant signal), not mere cell surface coating.

Discriminating controls: simulate_pulsed_duty_cycle_release + Trypan blue / cold-block / acid-wash / confocal validation ladder.